

Chapter:- AMINE



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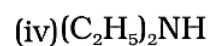
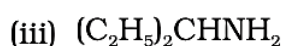
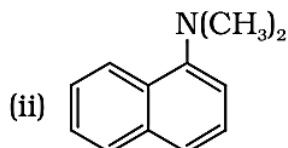
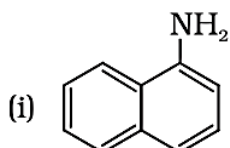
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Q1- Classify the following amines as primary, secondary or tertiary:



Q2- (i) Write structures of different isomeric amines corresponding to the molecular formula, $\text{C}_4\text{H}_{11}\text{N}$.

(ii) Write IUPAC names of all the isomers. (iii) What type of isomerism is exhibited by different pairs of amines?

Q3- Write chemical equations for the following reactions:

(i) Reaction of ethanolic NH_3 with $\text{C}_2\text{H}_5\text{Cl}$. (ii) Ammonolysis of benzyl chloride and reaction of amine so formed with two moles of CH_3Cl .

Q4- Write chemical equations for the following conversions:

(i) $\text{CH}_3-\text{CH}_2-\text{Cl}$ into $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{NH}_2$

(ii) $\text{C}_6\text{H}_5-\text{CH}_2-\text{Cl}$ into $\text{C}_6\text{H}_5-\text{CH}_2-\text{CH}_2-\text{NH}_2$

Q5- Write structures and IUPAC names of

(i) the amide which gives propanamine by Hoffmann bromamide reaction.

(ii) the amine produced by the Hoffmann degradation of benzamide.

Q6- How will you convert (i) Benzene into aniline (ii) Benzene into N, N-dimethylaniline (iii) $\text{Cl}-(\text{CH}_2)_4-\text{Cl}$ into hexan-1,6-diamine?

Q7- Arrange the following in decreasing order of their basic strength: $\text{C}_6\text{H}_5\text{NH}_2$, $\text{C}_2\text{H}_5\text{NH}_2$, $(\text{C}_2\text{H}_5)_2\text{NH}$, NH_3

Q8- Arrange the following in increasing order of their basic strength:

(i) $\text{C}_2\text{H}_5\text{NH}_2$, $\text{C}_6\text{H}_5\text{NH}_2$, NH_3 , $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$ and $(\text{C}_2\text{H}_5)_2\text{NH}$

(ii) $\text{C}_2\text{H}_5\text{NH}_2$, $(\text{C}_2\text{H}_5)_2\text{NH}$, $(\text{C}_2\text{H}_5)_3\text{N}$, $\text{C}_6\text{H}_5\text{NH}_2$

(iii) CH_3NH_2 , $(\text{CH}_3)_2\text{NH}$, $(\text{CH}_3)_3\text{N}$, $\text{C}_6\text{H}_5\text{NH}_2$, $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$.

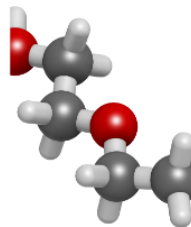
Q9- Complete the following acid-base reactions and name the products:

(i) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2 + \text{HCl} \rightarrow$

(ii) $(\text{C}_2\text{H}_5)_3\text{N} + \text{HCl} \rightarrow$

Q10- Write reactions of the final alkylation product of aniline with excess of methyl iodide in the presence of sodium carbonate solution.

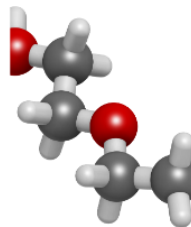
Q11- Write chemical reaction of aniline with benzoyl chloride and write the name of the product obtained.



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- Q12-** Write structures of different isomers corresponding to the molecular formula, C_3H_9N . Write IUPAC names of the isomers which will liberate nitrogen gas on treatment with nitrous acid.
- Q13-** How will you convert 4-nitrotoluene to 2-bromobenzoic acid ?
- Q14-** Convert
- (i) 3-Methylaniline into 3-nitrotoluene.
 - (ii) Aniline into 1,3,5 – tribromobenzene
- Q15-** Write IUPAC names of the following compounds and classify them into primary, secondary and tertiary amines
- (i) $(CH_3)_2CHNH_2$
 - (ii) $CH_3(CH_2)_2NH_2$
 - (iii) $CH_3NHCH(CH_3)_2$
 - (iv) $(CH_3)_3CNH_2$
 - (v) $C_6H_5NHCH_3$
 - (vi) $(CH_3CH_2)_2NCH_3$
 - (vii) $m\text{-}BrC_6H_4NH_2$
- Q16-** Give one chemical test to distinguish between the following pairs of compounds.
- (i) Methylamine and dimethylamine
 - (ii) Secondary and tertiary amines
 - (iii) Ethylamine and aniline
 - (iv) Aniline and benzylamine
 - (v) Aniline and N-methylaniline.
- Q17-** Account for the following:
- (i) pK_b of aniline is more than that of methylamine.
 - (ii) Ethylamine is soluble in water whereas aniline is not.
 - (iii) Methylamine in water reacts with ferric chloride to precipitate hydrated ferric oxide.
 - (iv) Although amino group is o– and p– directing in aromatic electrophilic substitution reactions, aniline on nitration gives a substantial amount of m-nitroaniline.
 - (v) Aniline does not undergo Friedel-Crafts reaction.
 - (vi) Diazonium salts of aromatic amines are more stable than those of aliphatic amines.
 - (vii) Gabriel phthalimide synthesis is preferred for synthesising primary amines.
- Q18-** Arrange the following:
- (i) In decreasing order of the pK_b values: $C_2H_5NH_2$, $C_6H_5NHCH_3$, $(C_2H_5)_2NH$ and $C_6H_5NH_2$
 - (ii) In increasing order of basic strength: $C_6H_5NH_2$, $C_6H_5N(CH_3)_2$, $(C_2H_5)_2NH$ and CH_3NH_2
 - (iii) In increasing order of basic strength:
 - (a) Aniline, p-nitroaniline and p-toluidine
 - (b) $C_6H_5NH_2$, $C_6H_5NHCH_3$, $C_6H_5CH_2NH_2$.
 - (iv) In decreasing order of basic strength in gas phase: $C_2H_5NH_2$, $(C_2H_5)_2NH$, $(C_2H_5)_3N$ and NH_3



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(v) In increasing order of boiling point: C_2H_5OH , $(CH_3)_2NH$, $C_2H_5NH_2$

(vi) In increasing order of solubility in water: $C_6H_5NH_2$, $(C_2H_5)_2NH$, $C_2H_5NH_2$.

Q19- How will you convert:

- (i) Ethanoic acid into methanamine
- (ii) Hexanenitrile into 1-aminopentane
- (iii) Methanol to ethanoic acid
- (iv) Ethanamine into methanamine
- (v) Ethanoic acid into propanoic acid
- (vi) Methanamine into ethanamine
- (vii) Nitromethane into dimethylamine
- (viii) Propanoic acid into ethanoic acid?

Describe a method for the identification of primary, secondary and tertiary amines. Also write chemical equations of the reactions involved.

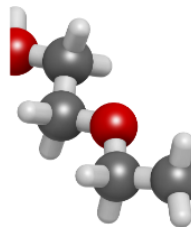
Q20- Write short notes on the following:

- (i) Carbylamine reaction
- (ii) Diazotisation
- (iii) Hofmann's bromamide reaction
- (iv) Coupling reaction
- (v) Ammonolysis
- (vi) Acetylation
- (vii) Gabriel phthalimide synthesis.

Q21- Accomplish the following conversions:

- (i) Nitrobenzene to benzoic acid
- (ii) Benzene to m-bromophenol
- (iii) Benzoic acid to aniline
- (iv) Aniline to 2,4,6-tribromofluorobenzene
- (v) Benzyl chloride to 2-phenylethanamine
- (vi) Chlorobenzene to p-chloroaniline
- (vii) Aniline to p-bromoaniline
- (viii) Benzamide to toluene
- (ix) Aniline to benzyl alcohol.

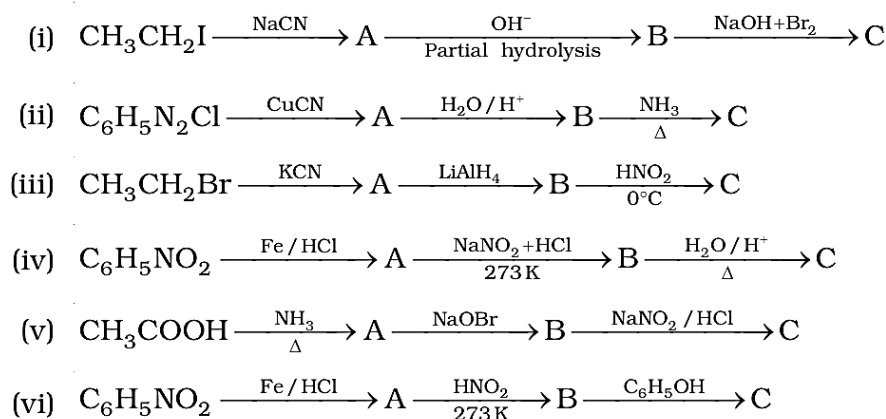
Q22- Give the structures of A, B and C in the following reactions:



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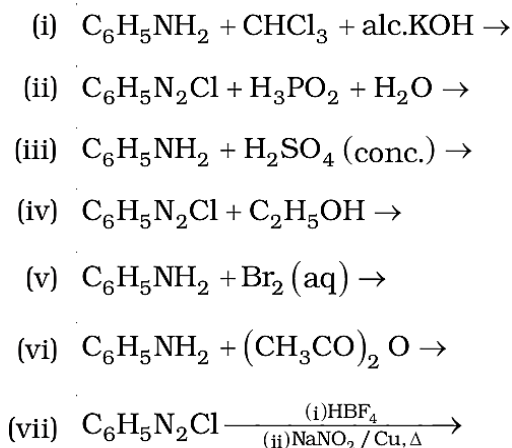
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Q23- An aromatic compound 'A' on treatment with aqueous ammonia and heating forms compound 'B' which on heating with Br₂ and KOH forms a compound 'C' of molecular formula C₆H₇N. Write the structures and IUPAC names of compounds A, B and C.

Q24- Complete the following reactions:



Q25- Why cannot aromatic primary amines be prepared by Gabriel phthalimide synthesis?

Q26- Write the reactions of (i) aromatic and (ii) aliphatic primary amines with nitrous acid.

Q27- Give plausible explanation for each of the following:

- Why are amines less acidic than alcohols of comparable molecular masses?
- Why do primary amines have higher boiling point than tertiary amines?
- Why are aliphatic amines stronger bases than aromatic amines?